



Restoration of foothills rough fescue grassland following pipeline disturbance in southwestern Alberta

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ABSTRACT

The effects of pipeline construction and reclamation techniques on the restoration of rough fescue plant communities following pipeline construction in southwestern Alberta, Canada were evaluated after 7–40 years. The pipeline construction right-of-way (ROW) sites varied from no recovery of rough fescue grassland to moderate recovery. The ROW sites had a higher proportion of introduced grasses and forbs, less topsoil, and poorer rangeland health than the adjacent undisturbed grassland. Within the ROW sites, less topsoil was present on those with larger diameter pipe and which had topsoil fully stripped from the ROW during construction. Introduced grasses, *Festuca ovina* (sheep fescue) and *Poa compressa* (Canada bluegrass), succeeded in establishment following seeding and persisted for at least 40 years. *Poa pratensis* (Kentucky bluegrass) dominated many of the ROW sites. Contributing factors to moderate recovery of rough fescue grassland were related to post-growing season pipeline construction, ideally, between August and March, summer or fall seeding, and minimum disturbance trench-only stripping. Reclamation practices appeared more important than time since restoration in the restoration of rough fescue grassland.

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1. Introduction

Fescue (*Festuca* spp.) grasslands are among the most threatened community types in the Northern Great Plains (Romo et al., 1990; Clark, 1998). Historically conversion to crop production was the main cause of change (Owens and Myers, 1973; Samson and Knopf, 1996). Other pressures on remnant fescue grasslands are country residential development, domestic livestock heavy grazing (Dormaar and Willms, 1990, 1998; Vujnovic et al., 2002), invasion by exotics and non-native species (Tyser and Key, 1988), tree and shrub invasion (Bailey and Wroe, 1974), and energy development. Fescue grasslands occur from southern British Columbia to western Ontario, in northeastern Washington Montana, North Dakota and at higher elevations in Colorado (Looman, 1982; Pavlick and Looman, 1984; Anderson, 2006). Since the onset of European settlement in the Alberta Foothills Fescue sub region, conversion to agriculture, urban or residential use has reduced the original fescue grassland (1.1 million hectares) by 83% (Adams et al., 2005). Tracts of native grassland remaining in the area are considered environmentally

significant for plant and animal biodiversity and cattle and ungulate forage value (Dormaar and Willms, 1990).

Festuca campestris is the dominant grass in the southwestern Alberta foothills, where it may grow almost to the exclusion of other plants once established (Looman, 1969; Willms et al., 1985, 1996). It is densely tufted, often forming large tussocks up to 30 cm in diameter, with stiff upright stems from 30 to 140 cm in height and roots up to 120 cm deep (Budd, 1987). The high curability of rough fescue permits winter grazing, reducing wintering costs, and making grazing options more flexible for the producer (Willms et al., 1996). Rough fescue (*F. campestris*) provides quality winter forage for elk and high cover values for a wide variety of wildlife species (Adams et al., 2005). Its canopy cover and deep rooting characteristics (Looman, 1969) result in little exposed soil and high soil stability, providing protection from weed invasion. Rough fescue communities produce substantial litter that serves to conserve scarce moisture and enhance moisture infiltration and retention, providing important contributions to the Alberta foothills watershed (Willms et al., 1988, 1996; Dormaar and Willms, 1990).

F. campestris is susceptible to human-caused disturbances. The effects of grazing on the reduction of *F. campestris* vigor and its eventual elimination from grassland vegetation have been documented (Willms, 1991; Willms et al., 1996; Dormaar and Willms, 1998). Little research has been conducted on restoring *F. campestris* following human disturbances.

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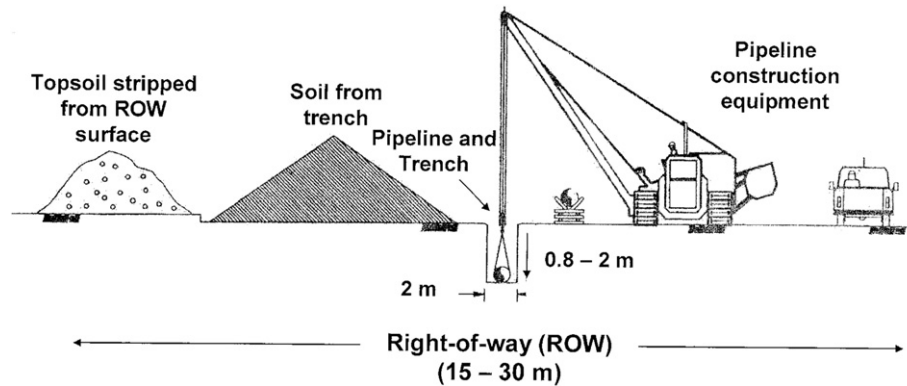


Fig. 1. Typical pipeline construction showing right-of-way (ROW), topsoil stripping and a trench in which the pipe is buried. ROW sampling plots were placed over the pipeline trench.

Pipeline construction prior to 1997 typically required a right-of-way (ROW) of 15–30 m in width from which the topsoil was stripped to accommodate the pipeline trench and construction equipment. Topsoil and subsoil were stripped and stockpiled for the duration of construction, usually less than one month (Fig. 1). Revegetation techniques following surface disturbance have changed over time. Little revegetation occurred prior to 1970. Up to the 1980s, seeding with imported grasses and cultivars such as *Festuca ovina* (sheep fescue) or *Poa pratensis* was a common reclamation and recovery practice in the oil and gas industry. Soil handling during pipeline trenching has evolved from simply replacing the soil with little concern for soil admixing to careful separation and replacement of the different soil horizons (Naeth et al., 1987).

Few references were found concerning invasion by non-native species into fescue grasslands. Revel (1993) transplanted rough fescue sod and reported a subsequent change in the transplanted plant community structure, away from dominance by deep-rooted fescue towards shallow-rooted rhizomatous grass species, e.g. *P. pratensis* (Kentucky bluegrass) or *Pascopyrum smithii* (western wheatgrass), and forbs. In Glacier National Park Montana, competitive agronomic grass species, e.g. *P. pratensis* and *Phleum pratensis* (timothy), invaded undisturbed fescue grasslands from roadsides and trails out to the most distant transects (100 m; Tyser and Worley, 1992). The dominance of non-native species following disturbance has been documented in pipeline reclamation studies in various grassland types in Alberta (Petherbridge, 2000; Elsinger, 2009). Clark (1998) concluded fescue grasslands are at more risk of conversion to non-native communities than other types of grassland in western Canada.

Grassland restoration is often unsuccessful as a consequence of ineffective reclamation techniques, unreliable seed sources, competition from weeds and agronomic species and variation in weather. The establishment time of rough fescue, the effect of grazing, and, the little understood ecology of associated forbs and grasses are some of the factors affecting disturbed rough fescue grasslands successful revegetation to their natural state.

In this study we examined pipeline construction methods, site conditions and revegetation methods on foothills fescue grassland plant community composition, invasion by non-native species, and shifts in species dominance along pipeline ROWs. Our objectives were to evaluate the effects of pipeline construction and reclamation techniques in rough fescue grassland and assess if revegetation resulted in the restoration of rough fescue plant communities.

2. Study area

The study area is located in the eastern foothills of the Rocky Mountains between latitudes 49.32° and 50.55° north and

longitudes 114.01° and 114.28° west, in the Foothills Fescue natural subregions of southwestern Alberta, Canada (Fig. 2). Mean annual precipitation varies from approximately 470 mm at lower altitudes to over 680 mm at higher altitudes, with approximately 20 cm of winter snow. Summers are cool, 15–18 °C mean daily temperatures and winters are cold, –10 to –5 °C, although occasionally warmed by the Chinook phenomenon, caused by catabatic winds off the Rocky Mountains (Strong and Leggat, 1992).

The criteria for choosing study sites were: 1) presence of one or more pipeline rights-of-way (ROW) that exhibited no major additional mechanical or grazing disturbance; 2) a known history of native seeding, including *F. campestris*, during reclamation; and 3) native grassland communities with *F. campestris* and associated grasses adjacent to the ROW serving as controls. Only pipelines constructed at least 7 years (and up to 40 years) prior were selected to allow for the establishment time of *F. campestris*, i.e., requiring up to 3–4 years to establish (Johnston and MacDonald, 1967). Approximately 25 pipeline ROWs in the study area met the seeding criteria, and of those only 14 met all the criteria. On all study sites topsoil had been stripped from the ROW with varying levels of disturbance.

3. Methods

3.1. Field sampling

Field sampling occurred between June 29 and August 08, 2004. We assessed the pipeline ROWs and the adjacent native grassland, with a stratified random sampling technique, at three topographic and hydrologic function locations: south-facing slopes (sub-mesic), crests (sub-xeric), and bottomland or north-facing slopes (mesic), all locations at least 100 m apart. We placed two 30-m line transects with 10 quadrats, at each ROW location: one over the pipeline trench and one in the adjacent native grassland as a control (Fig. 3). The reference plot

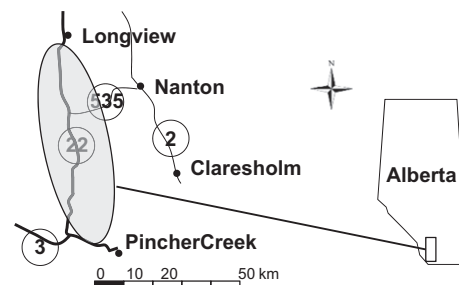


Fig. 2. Study area in southwestern Alberta, Canada. The Pipeline ROWs were located close to Highways 22 and 3, between Longview and Pincher Creek.

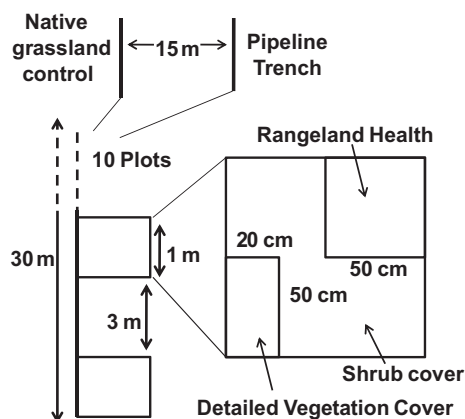


Fig. 3. Vegetation sampling method, showing the relative location of the ROW and control transects, the transect plot layout, and, the nested sampling plot design.

was placed approximately 15 m from the ROW in a similar original plant community not influenced by the ROW seeding. Nested subplots, at 3 m intervals, ten subplots per transect, were comprised of a 20 × 50 cm quadrat for detailed species percent cover estimates, a 50 × 50 cm frame for rangeland health, and a 1 × 1 m frame for shrub cover (Fig. 3). If another pipeline was situated in the same ROW and within 30 m of the first pipeline, it was also sampled, using the same reference plot for both pipelines. This resulted in a total of 56 pipeline ROW plots and 37 undisturbed grassland reference plots. All vegetative species were identified and their corresponding canopy cover and bare ground, were visually estimated (Anderson, 1986). Litter was hand-raked. On ROWs where *F. ovina* had been seeded, an additional assessment was made. Pin-point counts were made perpendicular to and from the edge of the ROW, until the last *F. ovina* plant was found. Nomenclature follows Kartesz (1994).

Slope was measured with a clinometer, aspect with a compass, and elevation with a global positioning system (GPS). Drainage was estimated from the site topographical position, e.g. crest, slope or bottomland. Topsoil depth was measured with pedons dug to a depth of 15 cm, limited to that depth by pipeline safety requirements. Soil texture was determined using manual texturing (Day, 1965; McKeague, 1978) and stoniness estimated following the Canadian System of Soil Classification: 0 = non-stony, 1 = slightly, 2 = moderately, 3 = very, and, 4 = exceedingly (Soil Classification Working Group, 1998).

Pipeline companies provided construction and reclamation reports with the following information: pipeline construction and reclamation dates, pipeline diameter (cm), seeding mixes and rates (kg/ha). Pipeline construction took place during all seasons. Pipe diameter varied from 11.4 to 107 cm, depths between 0.8 and 2 m to the top of the pipe. Site elevations were between 1250 and 1560 m above sea level. The age of pipelines ranged from seven to forty years. Seed mixes included harvested native seed or commercial mixes. *F. campestris* seeding rates varied from 1.3 to 12.5 kg/ha. Seeding occurred in spring, summer or fall and did not necessarily occur during the construction season.

3.2. Statistical methods

Dominance by *F. campestris* was considered “successful” restoration, while dominance by non-native species, e.g. *F. ovina* or *P. pratensis* was considered “failure” of rough fescue grassland restoration. We calculated mean cover, and frequency (the percent of quadrats in which a species occurred) for all plant species. The data were not normally distributed; therefore, we used Mann–Whitney *U*-tests to compare mean percent cover values between ROW and

controls and Kruskal–Wallis with Scheffé rank tests to determine differences among plant community groups and pipeline variables (Day and Quinn, 1989). We ran Wilcoxon signed ranks tests to assess the effect of seeding rates on resulting cover.

Non-metric multidimensional scaling (NMS) was used to ordinate and display the separation of native grassland controls to ROW species, and to distinguish ROW plots by floristic composition. The Sorenson (Bray–Curtis) distance measure was used to calculate the statistical significance of axes with a final run of 250 iterations using three axes and a randomly selected starting configuration. Individual species and pipeline and environmental variables were overlaid on the resulting ordination.

We performed indicator species analysis (ISA) to validate the dominant species of the plant communities derived from the NMS groupings. The indicator value (IV) ranges from 0 to 100, where 100 indicates a species is exclusively found in a particular group (Dufrene and Legendre, 1997). We grouped the ROW plots according to dominant vegetation species for further analyses. The nonparametric multiple response permutation procedure (MRPP), operating on Euclidean distances, was used to evaluate significant differences between ROW groups and controls (Zimmerman et al., 1985). MRPP generates a chance-corrected within-group agreement value (*A*), which evaluates the difference between species composition of grouped plots. The higher the *A* value, the more dissimilar are the groups (McCune and Grace, 2002).

We categorized the season in which the pipeline construction took place: Spring (April, May), Summer (June, July, August), Fall (September, October, November), and Winter (December, January, February, March). Month grouping were based on similar temperatures as they occurred in the region. We categorized the pipeline stripping method as four nominal classes: 20 – trench-only stripping, entailing a narrow disturbance of 2 m or less, the width of the trench; 50 – two-lift stripping, whereby the topsoil and subsoil are stripped separately over the full ROW width thus preserving the topsoil; 100 – full width stripping, when topsoil and subsoil are stripped and mixed together; and 150 – pipelines built before 1980 with little to no reclamation. For stepwise multiple regression, nominal classes were converted to binary coding (0 or 1) for each class (Quinn and Keough, 2002).

Topographical and pipeline construction variables were correlated with NMS axes using Pearson (2-tailed) correlation (quantitative variables) and Spearman's (2-tailed) rank correlation (categorical variables) to determine any effect of these variables on the floristic composition of the ROW plots. We evaluated the influence of pipeline variables on percent cover of dominant species using stepwise multiple regression. Statistical analyses were performed using SPSS (version 18) and PC-ORD (version 5.17).

We applied the rangeland health assessment protocol of Adams et al. (2003). Rangeland health is an aggregate measure of the departure of site conditions from a reference state for a plant community. It is based on scoring five variables:

- 1) Integrity and ecological status – is the plant community native or modified to non-natives species and, what is the successional status of the plant community?
- 2) Plant community structure – are the expected plant layers present or are any missing or significantly reduced?
- 3) Hydrologic function and nutrient cycling – are the expected amounts of organic residue present to safeguard hydrologic processes and nutrient cycling?
- 4) Site stability – is the site subject to accelerated erosion?
- 5) Noxious weeds – are noxious weeds present on the site?

The total score is converted to a percentage value, which is classified into one of three categories: unhealthy (0–49%), healthy with

problems (50–74%), healthy (75–100%). Reference states for plant communities in the study area were based on Adams et al. (2005).

4. Results

4.1. Pipeline ROW and native grassland controls

All introduced species had greater cover on ROWs than on control plots, e.g. *F. ovina*, *Poa compressa* (Canada bluegrass), *P. pratensis*, and weedy species such as *Taraxacum officinale* (dandelion) and *Cirsium arvense* (Canada thistle). Conversely, controls had greater cover by native forbs and graminoids overall than ROWs (Table 1), and *F. campestris* and *Danthonia parryi* significantly so. NMS ordination (stress level 17.6) showed separation of control and ROW plots with correlation of *F. campestris* and native forbs towards the controls. *P. pratensis*, weeds, *F. ovina*, and other introduced graminoids correlated towards the ROW plots (Fig. 4). *F. campestris* was present on twenty (45%) of the ROW plots, and where it occurred, its cover was 20–50% less than the control plots (Table 1). *P. pratensis* was the next most abundant species on both ROWs and controls, followed by *Bromus inermis*. Cover by native grasses *Elymus lanceolatus* (northern wheatgrass) and *P. smithii* was not significantly different between controls and ROWs (Table 1). On sites where *F. ovina* had been seeded, it was found to have spread between 3 and 9.7 m from the ROW edge. Less than 1% *F. ovina* cover occurred in the adjacent grassland controls, i.e., 15 m from the ROW edge.

Topsoil was deeper on the controls than on the ROWs ($p < 0.001$) and the ROWs had more bare soil ($p < 0.001$). Estimates of rangeland health were higher on control plots than ROW plots ($p < 0.001$). No significant differences were found for stoniness or the percent of sand and clay (Table 2).

NMS ordination resulted in a separation of ROW plots into three plant communities (stress level 13.96) (Fig. 5). We confirmed the dominant species in each community type with ISA: 18 plots were dominated by *F. ovina* (IV = 88, $p < 0.05$); 14 plots by *P. pratensis* (IV = 88, $p < 0.05$); and 10 plots co-dominated by *F. campestris* (IV = 56, $p < 0.05$) and *P. compressa* (IV = 58, $p < 0.05$). For further

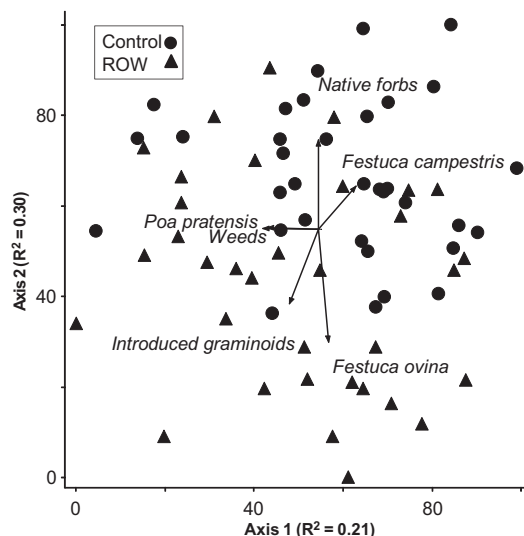


Fig. 4. Non-metric multidimensional scaling (NMS) analysis of pipeline ROW plots and native grassland controls, axes 1 and 2, showing separation between the majority of ROW and control plots, overlaid with vegetation response variables. Arrows on the NMS diagram show the strength and direction of a correlation, i.e., the longer the arrow, the stronger the correlation.

analyses, the ROW plots were grouped by and named for their dominant plant species (Table 3).

MRPP analysis showed the *F. ovina* ROW group was most different from control plots ($A = 0.07$, $p < 0.001$), the *P. pratensis* group was next ($A = 0.018$, $p = 0.056$), and the *F. campestris*/*P. compressa* group most resembled control plots ($A = 0.0006$, $p = 0.41$) (Table 3). Rangeland health was positively correlated with Axis 1 ($R^2 = 0.42$; $r^2 = 0.21$, $p = 0.003$) and Axis 3 ($R^2 = 0.21$; $r^2 = 0.13$, $p = 0.020$) in the NMS ordination corresponding to the *F. campestris*/*P. compressa* plant community (Fig. 5).

A few species showed significant differences based on moisture regime. *F. ovina* ($p = 0.005$) had greater cover on sub-xeric areas, *P. pratensis* ($p < 0.001$) was more abundant in mesic areas; however, *F. campestris* had no difference among three moisture regimes.

4.2. Pipeline construction

Pipeline construction season was positively correlated with Axis 3 ($R^2 = 0.21$; $r^2 = 0.17$, $p = 0.006$). Age of the ROW construction was not significantly correlated to either axes ($r^2 = 0.06$, $p = 0.124$ and $r^2 = 0.02$, $p = 0.394$). Among topographical and soil variables, only topsoil depth and percent clay were significantly correlated with the first two axes in the NMS ordination (Fig. 4). Topsoil depth ($r^2 = 0.18$, $p = 0.005$) was greatest in *F. campestris*/*P. compressa* communities and the highest clay content ($r^2 = 0.07$, $p = 0.047$) was present in the *F. ovina* and *P. pratensis* communities. Topsoil depth was negatively

Table 1

Vegetative species, with percent cover greater than 1%, and groups found on the controls and right-of-way (ROW) of pipeline construction showing mean percent cover (standard deviation) and frequency (F), as tested by Mann–Whitney U-tests and Indicator Species Analyses (IV).

	Controls			ROW			<i>p</i>
	(<i>n</i> = 37)	F	IV	(<i>n</i> = 56)	F	IV	
Total native species	71 (21)	100		39 (18)	100		<0.0001
Total shrubs	7 (11)	65		1.6 (3.6)	37		0.000
Total native graminoids	35 (17)	100		27 (15)	100		0.012
Total introduced graminoids	18 (14)	97		27 (14)	100		0.001
Total introduced species	21 (17)	97		32 (17)	100		0.001
Total native forbs	29 (11)	97		11 (12)	88		<0.0001
Shrubs							
<i>Rosa acicularis</i>	0.7 (2)	30	25	0.3 (1)	7	1	0.002
<i>Symphoricarpos occidentalis</i>	1.0 (3)	22	10	0.4 (1.6)	9	8	0.184
Native Graminoids							
<i>Danthonia parryi</i>	7 (9)	78	48	2 (5)	39	20	<0.0001
<i>Festuca campestris</i>	8.6 (10)	84	70	1.8 (3)	45	8	<0.0001
<i>Elymus lanceolatus</i>	8 (8)	89	46	6 (6)	82	42	0.281
<i>Pascopyrum smithii</i>	1 (3)	32	37	3.5 (6)	51	5	0.055
Introduced Graminoids							
<i>Poa compressa</i>	4 (9)	43	20	6 (11)	48	30	0.962
<i>Poa pratensis</i>	7 (12)	68	30	6 (9)	57	35	0.941
<i>Festuca ovina</i>	1.5 (4)	27	5	7 (10)	46	44	0.040

Table 2

Soil and range health properties found on the controls and right-of-way (ROW) of pipeline construction showing means (standard deviation), as tested by Mann–Whitney U-tests.

	Controls (<i>n</i> = 37)	ROW (<i>n</i> = 56)	<i>p</i>
Topsoil depth (cm)	16 (6)	8 (9)	<0.0001
Sand (%)	34 (19)	34 (17)	0.903
Clay (%)	32 (19)	33 (9)	0.939
Stoniness (1–3)	1.7 (1)	2.1 (1)	0.119
Bare Soil	11 (9)	26 (18)	<0.001
Range health (%)	69 (18)	43 (21)	<0.0001
Range health category	Healthy w problems	Unhealthy	

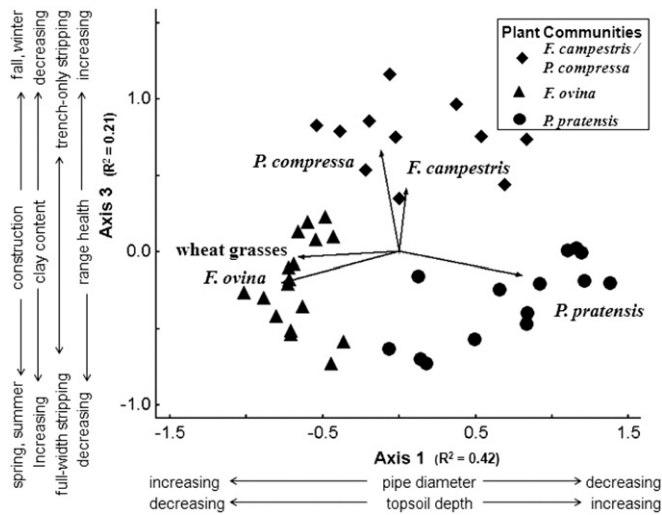


Fig. 5. Non-metric multidimensional scaling (NMS) analysis of pipeline ROW plots, axes 1 and 3, showing 3 plant communities overlaid with vegetation response variables and correlated with trends of pipeline factors. Arrows on the NMS diagram show the strength and direction of a correlation, i.e., the longer the arrow, the stronger the correlation.

correlated with pipe diameter ($r^2 = 0.26$, $p < 0.0001$). Among topographical and soil parameters included in stepwise multiple regression, only slope significantly influenced combined *F. campestris*/*D. parryi* cover (Cover = $6.98 - 0.40$ slope; $R^2 = 0.195$, $P = 0.003$); cover of these species was highest on gentle to moderate slopes. Rangeland health classifications for both the *F. ovina* and *P. pratensis* communities were classified as “unhealthy”, while the *F. campestris*/*P. compressa* community was “healthy with problems”.

Kruskal–Wallis tests showed associations similar to results of the NMS ordination overlays (Table 4). Pipe diameter was larger in *F. ovina*-dominated plots than the other two communities ($\chi^2 = 10.2$,

$p = 0.006$) and *F. ovina* cover was positively correlated with pipe diameter ($r^2 = 0.23$, $p = 0.001$). Pipe diameter was not significantly correlated with *F. campestris* cover ($r^2 = 0.03$, $p = 0.27$). Topsoil depth ($r^2 = 0.28$, $p < 0.0001$) and rangeland health ($r^2 = 0.18$, $p = 0.003$) were negatively correlated with pipe diameter.

The *F. ovina* community was associated with full width stripping and the other communities were associated with two-lift or trench-only methods ($\chi^2 = 10.1$, $p = 0.007$). *F. campestris*/*P. compressa* plant community plots were positively associated with pipeline construction in fall or winter ($\chi^2 = 10.0$, $p = 0.007$; Fig. 6). ‘Trench-only’ was the only stripping method found by stepwise multiple regression to significantly influence percent cover of *F. campestris*/*D. parryi* ($F = 15.6$, $P < 0.0001$). Similarly, the mean cover of *F. campestris*/*D. parryi* was significantly higher on trench-only ROW sites than on sites where other stripping methods were used ($\chi^2 = 9.80$, $p = 0.02$). In contrast, mean cover of *F. ovina* was least on trench-only ROW sites ($\chi^2 = 9.35$, $p = 0.025$; Fig. 7). Seeding rate significantly influenced percent cover of *F. ovina* ($r^2 = 0.10$, $p = 0.044$) but not *F. campestris* cover ($r^2 = 0.003$, $p = 0.72$).

5. Discussion

5.1. Re-vegetation effects

Reclamation of pipeline disturbances resulted in an altered state from original rough fescue grassland, declines in abundance and a number of native species, increases in abundance of introduced species, reduced plant diversity, altered soil conditions, and diminished rangeland health. *F. ovina* became established at most sites where seeded, and invaded adjacent sites and nearby grassland. Non-native species often dominate following disturbances such as pipelines (Petherbridge, 2000; Chen and Gao, 2006; Elsinger, 2009), roads (Tyser and Worley, 1992), ski hills (Muller et al., 1998) or cultivated land (Martin et al., 2005; Foster et al., 2007). As suggested by Muller et al. (1998) and Rhoady et al.

Table 3

Pipeline ROW plots grouped by dominant species showing mean percent cover (standard deviation) and frequency (F) with differences among groups tested by Kruskal–Wallis with Scheffé post hoc tests and Indicator Species Analyses (IV).

	ROW Plant Community Groups									p
	<i>F. ovina</i>			<i>P. pratensis</i>			<i>F. campestris</i> / <i>P. compressa</i>			
	(n = 18)	F	IV	(n = 14)	F	IV	(n = 10)	F	IV	
Total native species	41 (18)ab	100		36 (17)a	100		56 (18)b	100		0.015
Total shrubs	0.6 (2)a	17		3.0 (5)a	57		0.5 (1)a	40		0.082
Total native graminoids	9.4 (9)a	89		3.2 (5)a	86		9.6 (11)a	40		0.052
Total introduced graminoids	30 (16)a	100		23 (12)ab	100		12 (9)b	100		0.003
Total introduced species	33 (15)a	100		34 (13)a	100		16 (13)b	80		0.010
Total native forbs	6 (6)a	83		18 (13)b	100		17 (17)b	90		0.018
Shrubs								80		
<i>Rosa arkansana</i>	0a		0	1.8 (2)a	14	20	0.5 (2)a	25	1	0.619
<i>Symphoricarpos occidentalis</i>	0.2 (1)a	6	2	0.4 (1)a	21	13	0 (0)0a		0	0.220
Native Graminoids										
<i>Elymus lanceolatus</i>	9.4 (6)a	94	50	4.9 (5)a	93	23	5.1 (4)a	90	20	0.040
<i>Pascopyrum smithii</i>	6.6 (8)a	72	41	1.4 (3)b	43	7	0.3 (1)b	210	7	0.009
<i>Elymus trachycaulus</i>	1.3 (3)a	50	24	0.1 (0)a	14	1	1.2 (2)a	50	24	0.053
<i>Danthonia parryi</i>	0.6 (2)a	11	4	1.5 (3)a	50	9	5.5 (6)b	80	40	<0.0001
<i>Festuca campestris</i>	0.7 (2)a	33	6	0.8 (2)a	37	5	5.1 (4)b	90	56	0.001
<i>Festuca idahoensis</i>	3.1 (5)a	33	12	3.0 (4)a	43	14	3.0 (3)a	70	21	0.683
Introduced Graminoids										
<i>Bromus inermis</i>	1.4 (3)a	44	4	3.9 (4)a	64	36	1.3 (3)a	30	15	0.229
<i>Festuca ovina</i>	17 (10)a	89	88	2.2 (4)b	43	4	1.8 (4)b	20	0	<0.0001
<i>Phleum pratense</i>	3.8 (5)a	67	27	2.9 (5)a	57	20	1.0 (3)a	40	10	0.065
<i>Poa compressa</i>	7.1 (10)a	78	20	0.3 (1)b	14	0	9.9 (11)a	70	58	0.005
<i>Poa pratensis</i>	1.0 (2)a	33	2	17.6 (11)b	100	88	1.4 (1)a	70	3	<0.0001

Values followed by the same letter do not differ at $p > 0.01$ level. Only those with percent cover greater than 1.0% are shown.

Table 4

Pipeline construction, topographical, and, soil property variables, as grouped by ROW plant communities showing mean (standard deviation), tested by Kruskal–Wallis and Sheffé post hoc tests.

	ROW Plant Community Groups			
	<i>F. ovina</i> (n = 18)	<i>P. pratensis</i> (n = 14)	<i>F. campestris</i> / <i>P. compressa</i> (n = 10)	p
Pipeline age (years)	26 (17)a	17 (12)a	15 (11)a	0.227
Pipe diameter (cm)	100 (8)a	60 (39)b	54 (45)b	0.003
Stripping technique ¹	108 (43)a	59 (48)b	49 (49)b	0.002
Construction months ²	1.6 (1)a	2.8 (2)ab	3.9 (2)b	0.001
Sand (%)	36 (13)a	29 (11)a	43 (22)a	0.058
Clay (%)	30 (6)a	34 (2)a	28 (13)a	0.086
Ah depth (cm)	2.6 (4)a	8.9 (10)b	12 (7)b	0.001
Stoniness ³	2.1 (1)a	2.3 (1)a	2.1 (1)a	0.525
Aspect °	154 (118)a	272 (102)b	203 (122)ab	0.042
Slope (%)	9 (7)a	8 (5)a	5 (6)a	0.195
Elevation (m)	1424 (69)a	1354 (112)a	1400 (97)a	0.081
Litter (kg/ha)	112 (95)a	215 (255)a	131 (89)a	0.475

Values followed by the same letter do not differ at $p > 0.05$.

(1997), original seed bank and adjacent seed rain sources may not be sufficient to return original species once applied seed mix species have become established.

F. ovina, native to Europe, was included in the seed mix of all the assessed pipelines reclaimed prior to 1997. It is readily available as a cultivar and was widely included in early reclamation seed mixes. This grass is well adapted to biophysical conditions present in southwestern Alberta (Tannas, 2001). *F. ovina* has a shallow root system, requires little water, is drought and grazing tolerant, and grows well on nutrient-deficient soils (Brar and Palazzo, 1995; Otsus and Zobel, 2004; Hartley and Mitchell, 2005). While *F. campestris* is also tolerant of grazing and dry conditions (Willms et al., 1985), it is slower to establish, requiring at least 3–4 years for stand development. *F. ovina* sets seed in the first and every subsequent year, while *F. campestris* may not set seed for the first several years and then only every 5 to ten years (Johnston and MacDonald, 1967). The poor establishment of *F. campestris* may be related to its inability to compete with faster establishing *F. ovina*.

One non-native grass that was not seeded, but, nonetheless well-established on many plots, was *P. pratensis*. This species is an exotic in North America, presumably arriving with European settlement (Tannas, 2001). It is now naturalized in Alberta and

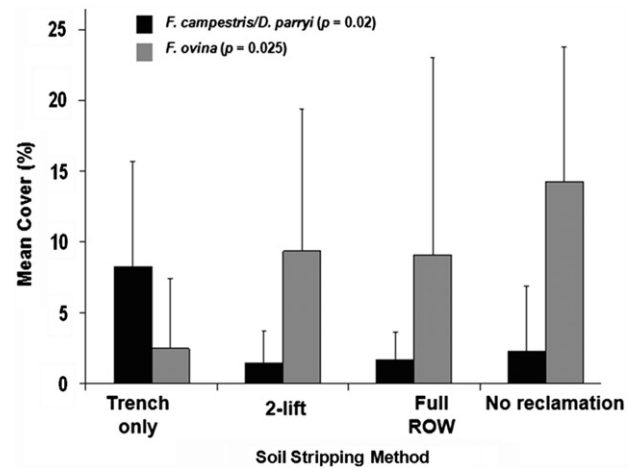


Fig. 7. Comparison of the variation in cover of *Festuca campestris/D. parryi* and *Festuca ovina* as a result of the amount of soil stripping used during pipeline construction, analyzed by Kruskal–Wallis means comparison. Error bars represent standard deviation.

often establishes from the seed bank when soil is disturbed (Adams et al., 2005). Unlike *F. campestris*, *P. pratensis* readily produces seed in its first and subsequent years, and is strongly rhizomatous, allowing it to establish and spread rapidly. We found very little *F. campestris* on pipelines where *P. pratensis* was dominant, suggesting the competitive strength of *P. pratensis*. *P. compressa*, another European native species, was included in the seed mix of most of the pipelines and persisted following reclamation seeding. Where *F. ovina* was not present, *F. campestris* was able to establish and co-dominate with *P. compressa*. While similar morphologically to *P. pratensis*, *P. compressa* is not considered as invasive (Tannas, 2001) and may not have competed as strongly against *F. campestris*.

5.2. Pipeline construction effects

Pipeline construction affects soil characteristics. Soil stripping may reduce organic matter and increase clay content through admixing of topsoil with subsoil (Culley et al., 1982; Naeth et al., 1987). Topsoil may be lost or degraded due to wind or water erosion soon after it is returned to the ROW (Naeth et al., 1987). The amount of organic matter and clay in soil influence vegetative growth. *F. ovina* was most abundant on sites with the highest degree of topsoil loss, in particular the larger diameter pipelines. The lower nutrient capacity of a shallower topsoil layer and the reduced moisture retention capacity of the altered ROW were conditions that appeared to favor *F. ovina*.

F. campestris recovery suffered least from trench-only stripping, the lowest impact technique documented in this study, resulting in a narrow disturbance (the width of the soil excavator: 30–60 cm). This construction method presumably allowed native grassland, in close proximity on either side of the disturbance, to promote recovery of *F. campestris* through seed rain and tillering from older plants close to the disturbance (Maki, 1989; Sinton, 2001).

Grassland recovery in our study was also influenced by construction season. *F. campestris* mainly became established at minimally disturbed sites on pipelines constructed in late fall or winter. During the winter, heavy equipment operating on frozen grassland results in limited disturbance and requires less topsoil stripping than in other seasons (Sinton, 2001). Similarly, Wilson (1988) concluded that military tank traffic in the late summer post-growing season had a less detrimental effect on prairie vegetation than traffic during the spring growing season.

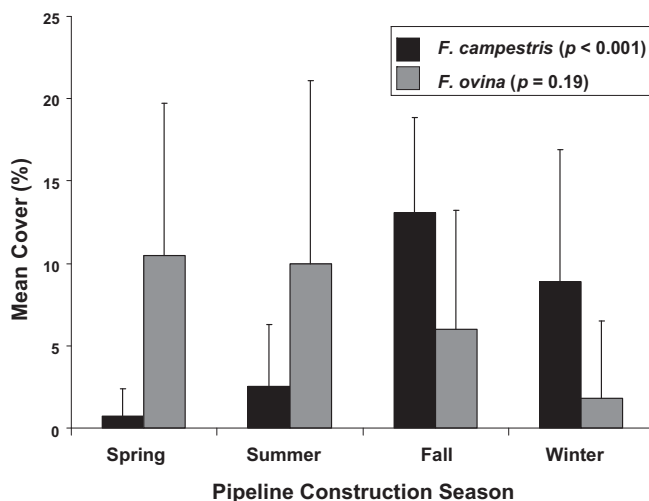


Fig. 6. Comparison of the variation in cover of *Festuca campestris* and *Festuca ovina* as a result of the season of pipeline construction, analyzed by Kruskal–Wallis means comparison. Error bars represent standard deviation.

F. campestris seed used on pipelines investigated in our study was wild-harvested. The germination rates of wild-harvested seeds of this species are usually lower and more variable than those from cultivated plants (Smreciu et al., 2003). Since *F. campestris* seed in this study was likely wild-harvested, this may have contributed to the poor establishment of *F. campestris* and the lack of relationship between *F. campestris* cover and its seeding rate. Contrary to the accepted practice of spring seeding for rough fescue (Pahl and Smreciu, 1999), our analysis showed that *F. campestris* succeeded better with late season seeding, i.e., late summer or fall. The literature contains little consensus regarding seeding rates for native species. Recommended seeding rates range from 4 to 8 kg/ha (240–480 seeds/m²) for rough fescue in Alberta (Pahl and Smreciu, 1999), up to 500 kg/ha (12,500 seeds/m²) for intermediate wheatgrass to improve competition with weedy species in Montana (Sheley et al., 2005), or 48 kg/ha (1160 seeds/m²) for western wheatgrass establishment in Kansas (Launchbaugh and Owensby, 1970).

We found no significant relationship between the age of the pipeline and the degree of recovery. Furthermore, older pipeline ROWs seeded with exotics remained dominated by exotics. In contrast, minimal disturbance and seeding techniques used on more recent pipelines produced variations resembling the original grasslands, with ROWs retaining many of the original species. Ecological resilience theory suggests that alternative succession pathways and stable states are possible following disturbance (Briske et al., 2008). In this study, disturbances associated with older pipelines (constructed prior to 1997) appeared to have triggered a change in the trajectory of original grassland toward a different stable state. Pipelines constructed with minimal disturbance techniques resulted in vegetation cover possibly continuing on a trajectory back toward the original state.

6. Conclusion

Mitigation of industrial disturbances on fescue grassland communities is difficult and evidence shows that post-disturbance restoration was not achieved with the technology and practices used at the time pipelines were constructed. Since then, minimum disturbance techniques have been developed and should be evaluated over time. Further study is required on seeding rate and type of rough fescue seed, and on other techniques, such as transplants or pre-grown plugs. Based on this study, we recommend the following practices: pipeline construction in fescue grasslands should take place in dry or frozen conditions, i.e., post-growing season; soil admixing should be avoided; disturbance width should be minimized; and seed mixes should not include non-native species.

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References

Adams, B.W., Ehler, G., Stone, C., Lawrence, D., Alexander, M., Willoughby, M., Hincz, C., Moisey, D., Bogen, A., 2003. Rangeland Health Assessment for Grassland, Forest and Tame Pasture (Pub. No. T/044). Public Lands and Forest Division, Alberta Sustainable Resource Development, Lethbridge, AB.

Adams, B.W., Ehler, R., Moisey, D., McNeil, R., 2005. Rangeland Plant Communities and Range Health Assessment Guidelines for the Foothills Fescue Natural Subregion of Alberta (Pub. No. T/044). Rangeland Management Branch, Public Lands Division, Alberta Sustainable Resource Development, Lethbridge, AB.

Anderson, E.W., 1986. A method for estimating cover. *Rangelands* 8, 236–238.

Anderson, G.D., 2006. *Festuca campestris* Rydberg (Rough Fescue): a Technical Conservation Assessment. USDA Forest Service, Rocky Mountain Region. Colorado State University, Fort Collins, CO.

Bailey, D.J., Wroe, R.A., 1974. Aspen invasion in a portion of the Alberta parklands. *Journal of Range Management* 27, 263–266.

Brar, G.S., Palazzo, A.J., 1995. Tall and hard fescues responses to periodic soil water deficits. *Journal of Agronomy and Crop Science* 175, 221–229.

Briske, D.D., Bestelmeyer, B.T., Stringham, T.K., Shaver, P.L., 2008. Recommendations for development of resilience-based state-and-transition models. *Rangeland Ecology and Management* 61, 359–367.

Budd, A.C., 1987. Budd's Flora of the Canadian Prairie Provinces. Canadian Government Publishing Centre, Hull, PQ.

Chen, L., Gao, Q., 2006. Chance and challenge for China on ecosystem management: lessons from the west-to-east pipeline project construction. *Ambio* 35, 91–93.

Clark, G.T., 1998. Fescue grassland restoration: integrating research and experience into a fescue grassland conservation strategy. In: *Proceedings of the Fifth Prairie Conservation and Endangered Species Conference*, pp. 61–65.

Culley, J.L.B., Dow, B.K., Presant, E.W., MacLean, A.J., 1982. Recovery of productivity of Ontario soils disturbed by an oil pipeline installation. *Canadian Journal of Soil Science* 62, 267–279.

Day, P.R., 1965. Particle fractionation and particle size analysis. In: Black, C.A. (Ed.), *Methods of Soil Analysis*, Part I. Soil Science Society of America, Madison, Wisconsin, pp. 457–567.

Day, R.W., Quinn, G.P., 1989. Comparison of treatments after an analysis of variance in ecology. *Ecological Monographs* 59, 433–463.

Dormaar, J.F., Willms, W.D., 1990. Sustainable production from the rough fescue prairie. *Journal of Soil and Water Conservation* 45, 137–140.

Dormaar, J.F., Willms, W.D., 1998. Effect of forty-four years of grazing on fescue grassland soils. *Journal of Range Management* 51, 122–126.

Dufrene, M., Legendre, P., 1997. Species assemblages and indicator species: the need for a flexible asymmetrical approach. *Ecological Monographs* 67, 345–366.

Elsinger, M., 2009. Reclamation Status of Plains Rough Fescue Grasslands at Rumsey Block in Central Alberta After Oil and Gas Well Site and Pipeline Disturbance. Thesis, University of Alberta, Edmonton, AB.

Foster, B.L., Murphy, C.A., Keller, K.R., Aschenbach, T.A., Questad, E.J., Kindscher, K., 2007. Restoration of prairie community structure and ecosystem function in an abandoned hayfield: a sowing experiment. *Restoration Ecology* 15, 652–661.

Hartley, S.E., Mitchell, R.J., 2005. Manipulation of nutrients and grazing levels on heather moorland: changes in *Calluna* dominance and consequences for community composition. *Journal of Ecology* 93, 990–1004.

Johnston, A., MacDonald, M.D., 1967. Floral initiation and seed production in *Festuca Scabrella* Torr. *Canadian Journal of Plant Science* 47, 577–583.

Kartesz, J.T., 1994. A Synonymized Checklist of the Vascular Flora of the United States, Canada and Greenland. In: *Checklist*, vol. 1. Timber Press, Portland, OR.

Launchbaugh, J.L., Owensby, C.E., 1970. Seeding rate and first-year stand relationships for six native grasses. *Journal of Range Management* 6, 414–417.

Looman, J., 1969. The fescue grasslands of Western Canada. *Vegetatio* 19, 128–145.

Looman, J., 1982. Grasslands of Western North America: fescue grasslands. In: *Grassland Ecology and Classification Symposium*.

Maki, D.R., 1989. The directly ploughed concept for small-inch pipelines – a modern approach to reducing installed costs and minimizing environmental impact. *Pipeline Technology* 28, 124–129.

Martin, L.M., Moloney, K.A., Wilsey, B.J., 2005. An assessment of grassland restoration success using species diversity components. *Journal of Applied Ecology* 42, 327–336.

McCune, B., Grace, J.B., 2002. *Analysis of Ecological Communities*. MjM Software, Gleneden Beach, OR.

McKeague, J.A., 1978. *Manual on Soil Sampling and Methods of Soil Analysis*. Canadian Society of Soil Sciences, Ottawa, ON.

Muller, S., Dutoit, T., Alard, D., Gréville, F., 1998. Restoration and rehabilitation of species-rich grassland ecosystems in France: a review. *Restoration Ecology* 6, 94–101.

Naeth, M.A., McGill, W.B., Bailey, A.W., 1987. Persistence of changes in selected soil chemical and physical properties after pipeline installation in solonchic native rangeland. *Canadian Journal of Soil Science* 67, 747–763.

Otsus, M., Zobel, M., 2004. Moisture condition and the presence of bryophytes determine fescue species abundance in a dry calcareous grassland. *Oecologia* 138, 293–299.

Owens, R.A., Myers, M.T., 1973. Effects of agriculture upon populations of native passerine birds of an Alberta grassland. *Canadian Journal of Zoology* 51, 697–713.

Pahl, M.D., Smreciu, A., 1999. *Growing Native Plants of Western Canada: Common Grasses and Wildflowers*. Alberta Agriculture, Food and Rural Development, Edmonton, AB.

Pavlick, L.E., Looman, J., 1984. Taxonomy and nomenclature of rough fescues, *Festuca altaica*, *F. campestris* (*F. scabrella* var. *major*), and *F. hallii*, in Canada and the adjacent part of the United States. *Canadian Journal of Botany* 62, 1739–1749.

Petherbridge, W.L., 2000. *Sod Salvage and Minimal Disturbance Pipeline Restoration Techniques: Implications for Native Prairie Restoration*. Thesis, University of Alberta, Edmonton, AB.

- Quinn, G.P., Keough, M.J., 2002. *Experimental Design and Data Analysis for Biologists*. Cambridge University Press, Cambridge, UK.
- Revel, R.D., 1993. Canada's rough fescue grasslands: a trial restoration is yielding encouraging results. *Restoration and Management Notes* 11, 117–124.
- Rhoady, B.A., Shaw, N.L., Booth, D.T., 1997. Using native seeds on rangelands. In: *Proceedings: Using Seeds of Native Species on Rangelands*, pp. 1–8.
- Romo, J.T., Grilz, P.L., Driver, E.A., 1990. Invasion of the Canadian prairies by an exotic perennial. *Blue Jay* 48, 130–135.
- Samson, F.B., Knopf, F.L., 1996. *Prairie Conservation: Preserving North America's Most Endangered Ecosystem*. Island Press, Washington, DC.
- Sheley, R.L., Jacobs, J.S., Svejcar, T.J., 2005. Integrating disturbance and colonization during rehabilitation of invasive weed-dominated grasslands. *Weed Science* 53, 307–314.
- Sinton, H.M., 2001. *Oil and Gas: a Lighter Footprint*. Alberta Environment, Edmonton, AB.
- Smreciu, E.A., Sinton, H.M., Walker, D.G., Bietz, J.K., 2003. *Establishing Native Plant Communities*. Alberta Agriculture, Food and Rural Development, Alberta Environment and Alberta Sustainable Resource Development, Edmonton, AB.
- Soil Classification Working Group, 1998. *The Canadian System of Soil Classification*. Agriculture and Agri-Food Canada Publication 1646, Ottawa, ON.
- Strong, W.L., Leggat, K.R., 1992. *Ecoregions of Alberta*. Alberta Forestry, Lands and Wildlife, Edmonton, AB.
- Tannas, K., 2001. *Common Plants of the Western Rangeland*. In: *Grasses and Grass-like Species*, vol. 1. Alberta Agriculture, Food and Rural Development, Edmonton, AB.
- Tyser, R., Worley, C., 1992. Alien flora in grasslands adjacent to road and trail corridors in Glacier National Park, Montana. *Conservation Biology* 6, 253–261.
- Tyser, R.W., Key, C.H., 1988. Spotted knapweed in natural area fescue grasslands: an ecological assessment. *Northwest Science* 62, 151–160.
- Vujnovic, K., Wein, R.W., Dale, M.R.T., 2002. Predicting plant species diversity in response to disturbance magnitude in grassland remnants of central Alberta. *Canadian Journal of Botany* 80, 504–511.
- Willms, W.D., 1991. Cutting frequency and cutting height effects on rough fescue and Parry oat grass yields. *Journal of Range Management* 44, 82–86.
- Willms, W.D., Adams, B.W., Dormaar, J.F., 1996. Seasonal changes of herbage biomass on the fescue prairie. *Journal of Range Management* 49, 100–104.
- Willms, W.D., Dormaar, J.F., Schaalje, G.B., 1988. Stability of grazed patches on rough fescue grasslands. *Journal of Range Management* 41, 503–508.
- Willms, W.D., Smoliak, S., Dormaar, J.F., 1985. Effects of stocking rate on a rough fescue grassland. *Journal of Range Management* 38, 220–225.
- Wilson, S.D., 1988. The effects of military tank traffic on prairie: a management model. *Environmental Management* 12, 397–403.
- Zimmerman, G.M., Goetz, H., Mielke, P.W.J., 1985. Use of an improved statistical method for group comparisons to study effects of prairie fire. *Ecology* 66, 606–611.